Lecture 5 (Accuracy & Confusion Matrix)

To check whether the performance of a classification matrix we have some classification matrix just like regression matrix for regression algorithms.

Accuracy

Suppose we have a data

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1000

800   200

LOR   DT

Actual	LOR	DT
1	1	1
0	1	1
0	0	0
0	0	0
1	1	1
1	1	1
0	1	0
0	0	0
0	0	0
1	1	1

this how we check our model accuracy

$$acc = \frac{\text{no of correct}}{\text{total no prediction}}$$

80%   90%

DT better

How to implement it

Cellwise code

Call data

df.head (3)

from sklear model selection import train test split

sklear liner model import logistic

. tree import DT

clf 1 = LR() Logistic regression

clf 2 = DTC() Desicision tree Classifier()

clf1. fit (X train, Y train)

clf2. fit (X train, Ytrain)

from sk learn metrics. import accuracy score, Confusion matrix

print ("Accuracy score LR, accu score (ytest, ypred)

·DT,

from sklearn metrics import classification report

now here we have binary

means either 0 or 1

Multiclassification data   Flower data
0   1   2

Same.

Actual   LR   DT   Same

accuracy = All correct prediction / total no. of prediction

Now Some talk on Accuracy

⇒ How Much Accuracy is Good?

The Higher ones X

It depends on data we are working on!

The Problem with Accuracy

Accuracy tells us

90%   10%   ? Correct Still missing

Y → X   Which
N → Yes.   Can't tell

To Rectify this thing we have
Confusion Matrix

from sklearn.metrics import accuracy_score, Confusion matrix
Confusion_matrix (y_test)

Confusion Matrix

	1 Prediction	0 Prediction
Actual 1	TP 26	FN actual 6
0	FP 0	TN 29

So Confusion matrix predicts types too.

accuracy can be predicted or calculated but not vice versa $= \dfrac{TP+TN}{TP+TN+FP+FN}$

Type1 Error : No of false positive

Type2 Error : No of false negative

How Confusion Matrix looks for multi class problem with eg (0, 1, 2)

jitni class utna by ka matrix

ie 3 class to 3X3 ka

	Predicted 0	Predicted 1	Predicted 2
actual	7		
		21	
			13

accuracy = total di / total

Similar 10x10 any kind

When Accuracy is Misleading?

In case of Imbalanced data set

Yes	No
900	100

Ex Airport Passenger terrorist

Accuracy	Predicted 1	Predicted 0
1	0	1
0	0	99999

mean 99.50%

wrong accuracy.